LED Color Growth Experiment — Student Worksheet

Artificial Light Growth Experiment and Lunar Growth Exploration

Name: **Date:**

Part 1 — Background

1. What do plants need to survive? List at least three things.

2. What is photosynthesis? Write a brief definition in your own words.

3. Why might growing plants on the moon be a challenge for astronauts? List two challenges.

Part 2 — Hypothesis

Complete the hypothesis statement below:

Tip: think about what you know about the light spectrum and photosynthesis.

If plants are grown under _____ colored light, then they will grow _____ than plants under white light,

because _____.

Independent variable (what you change): Dependent variable (what you measure):

List two variables you will keep the same (controlled variables):

Part 3 — Data Collection

Record plant height (cm) and observations each session. After germination, thin to the tallest seedling and note the date.

Date	Day #	Height (cm)	Qualitative observations (color, leaves, health)	Watered? ✓
Setup				
Week 1 – Mon				
Week 1 – Wed				
Week 1 – Fri				
Week 2 – Mon				
Week 2 – Wed				
Week 2 – Fri				
Week 3 – Mon				
Week 3 – Wed				
Week 3 – Fri				

Germination Date (when first sprout appeared):

Final Height (in cm):

Part 4 — Class Data

Record the class averages for each light color from the board:

Light color	Average final height (cm)	Notable observations
White (control)		
Red		
Blue		
Green		
Yellow		

4. Which light color produced the most growth? Which produced the least?

5. Did the class results match your hypothesis? Explain.

Part 5 — Claim, Evidence, Reasoning (CER)

Write a CER paragraph summarizing what you learned about light color and plant growth.

Claim — what did you find?

Evidence — what data supports it?

Reasoning — why does the evidence support the claim?

Part 6 — What If?

Choose one question and answer it below:

A) What if astronauts could only bring one type of LED light to the moon? Based on your findings, which would you recommend and why?

B) What if you varied light brightness instead of color? Predict what might happen and briefly design an experiment.

C) What if you had to calculate the cost and weight of a lunar greenhouse lighting system? What tradeoffs would you consider?

I chose question: ___